

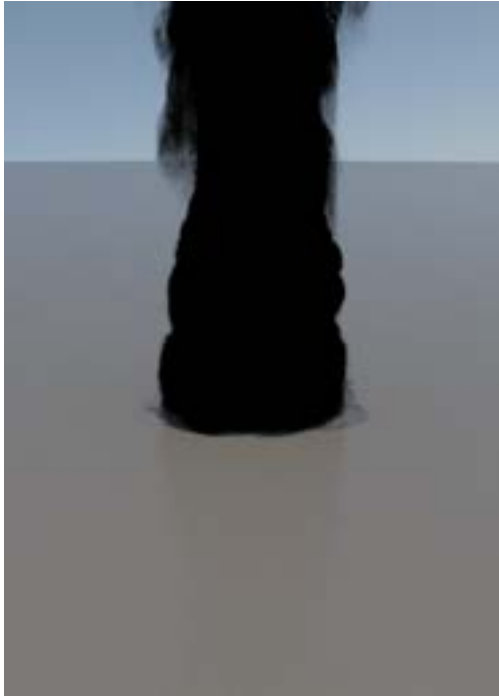
Appendix

Inputs	Outputs
Ray Origin x, y, z	Volumetric emitted radiance $L_{\text{intrinsic}}$
Ray Direction x, y, z	Weighted Throughput Ratio T
tMax parametric value until ray intersection	

Table 1: Instant-NGP model Inputs and Outputs



View 1



View 2



View 3

Figure 1: Inference for Native PBRT path on 3 determined novel views



View 1



View 2

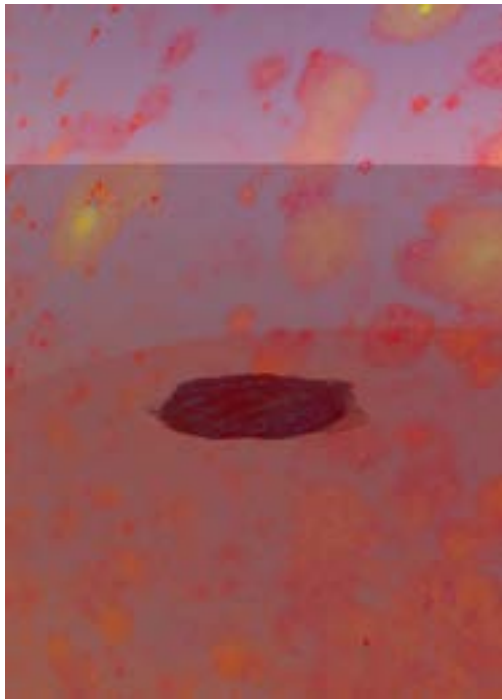


View 3

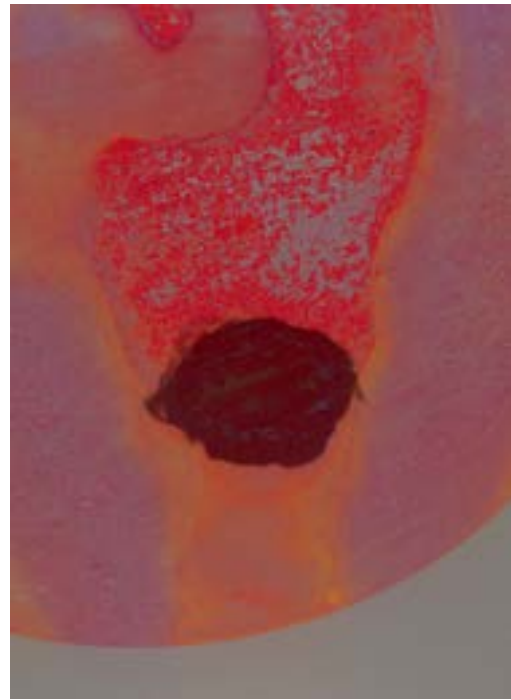
Figure 2: Inference for Spectral on 3 determined novel views



View 1



View 2



View 3

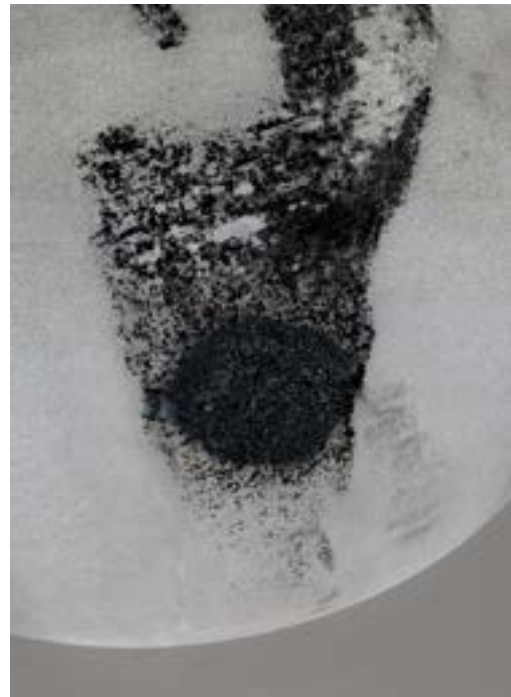
Figure 3: Inference for 128 neurons 3 layers color on 3 determined novel views



View 1

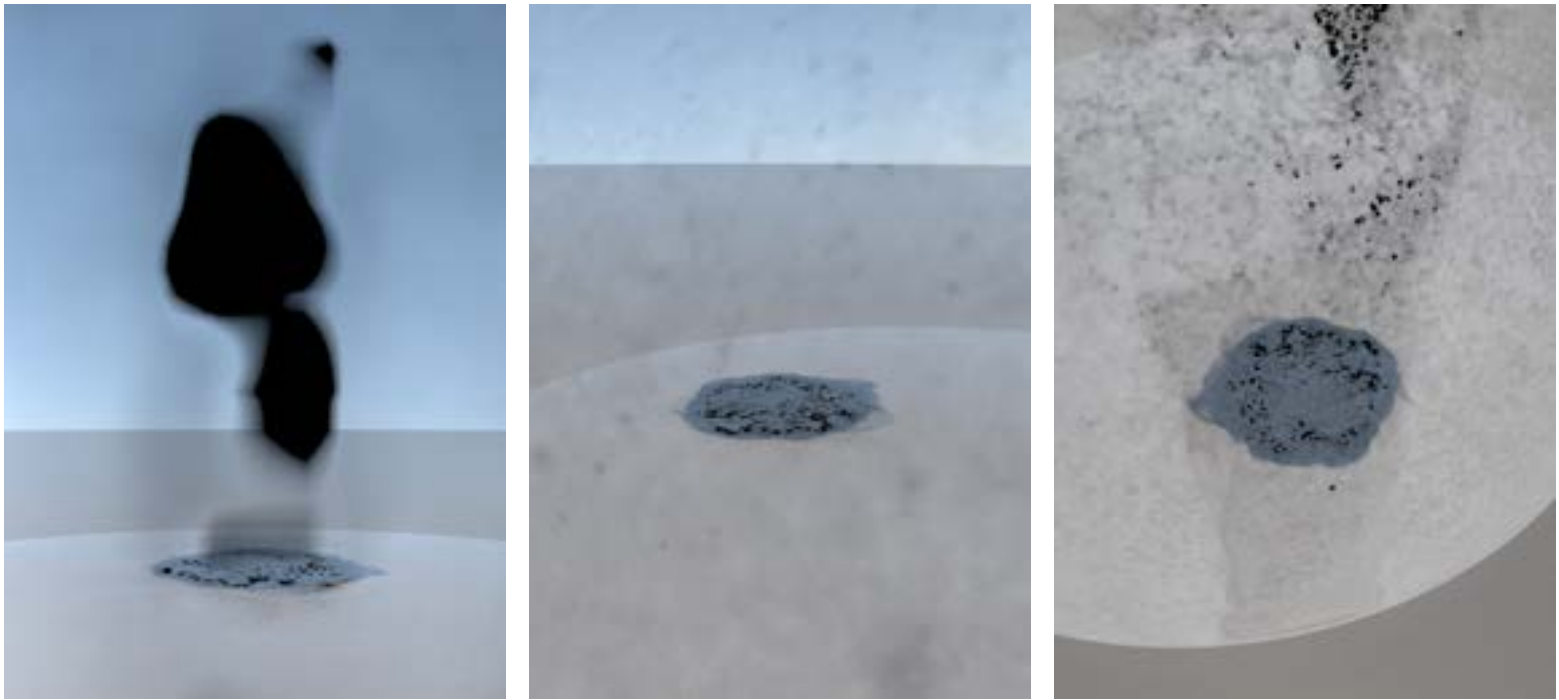


View 2



View 3

Figure 4: Inference for 128 neurons 3 layers T only on 3 determined novel views

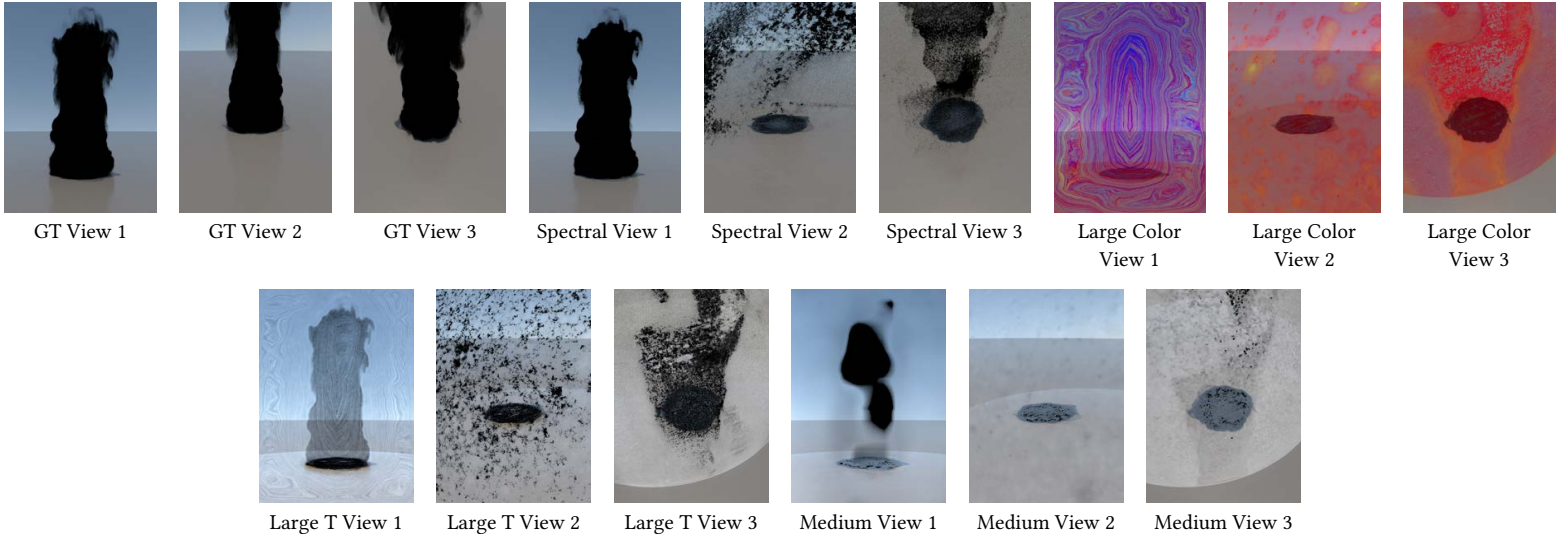


View 1

View 2

View 3

Figure 5: Inference for 12 nlevels and log hashgrid size 19 on 3 determined novel views



GT View 1

GT View 2

GT View 3

Spectral View 1

Spectral View 2

Spectral View 3

Large Color View 1

Large Color View 2

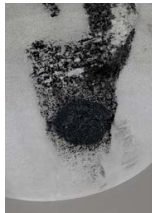
Large Color View 3



Large T View 1



Large T View 2



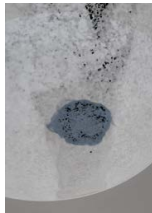
Large T View 3



Medium View 1



Medium View 2



Medium View 3

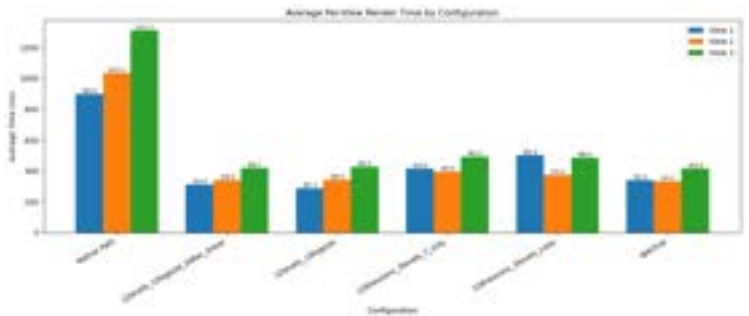


Figure 8: Graph displaying average time for 3 determined novel views across 3 runs.

Network Configuration	Hashgrid Size	MLP Size	Staging Buffers (e.g. inferBefore)	tcnn activation memory	Total
Medium.json	25,165,824 bytes $\approx$ 24MB	$13,312 \times 2 = 26,624$ bytes $\approx$ 26KB	$780,032 \times 188 = 146,646,016$ bytes $\approx$ 147 MB	$(3 \times 64 + 48) \times 2 \times 780032 = 374,415,360$ bytes $\approx$ 374 MB (with 75 MB input staging buffer)	546,253,824 bytes $\approx$ 546 MB
Large.json (both T and color)	33,554,432 bytes $\approx$ 32 MB	43008 bytes $\times 2 \approx$ 86KB	$780,032 \times 188 = 146,646,016$ bytes $\approx$ 147 MB	$(3 \times 128 \times + 64) 2 \times 780032 = 698,908,672$ bytes $\approx$ 699 MB (with 100MB input staging buffer)	879,195,136 bytes $\approx$ 879MB
Spectral.json	33,554,432 bytes $\approx$ 32 MB	43008 bytes $\times 2 \approx$ 86KB	$780,032 \times 188 = 146,646,016$ bytes $\approx$ 147 MB	$(3 \times 128 \times + 64) 2 \times 780032 = 698,908,672$ bytes $\approx$ 699 MB (with 100MB input staging buffer)	879,195,136 bytes $\approx$ 879 MB (less if optimized to exclude certain staging buffers)

Table 2: Table showing memory usage per configuration. Usage is split between Network weight size and Staging Buffer size with JIT Fusion disabled. If JIT Fusion is enabled, the tcnn staging memory drastically decreases and only accounts for input size

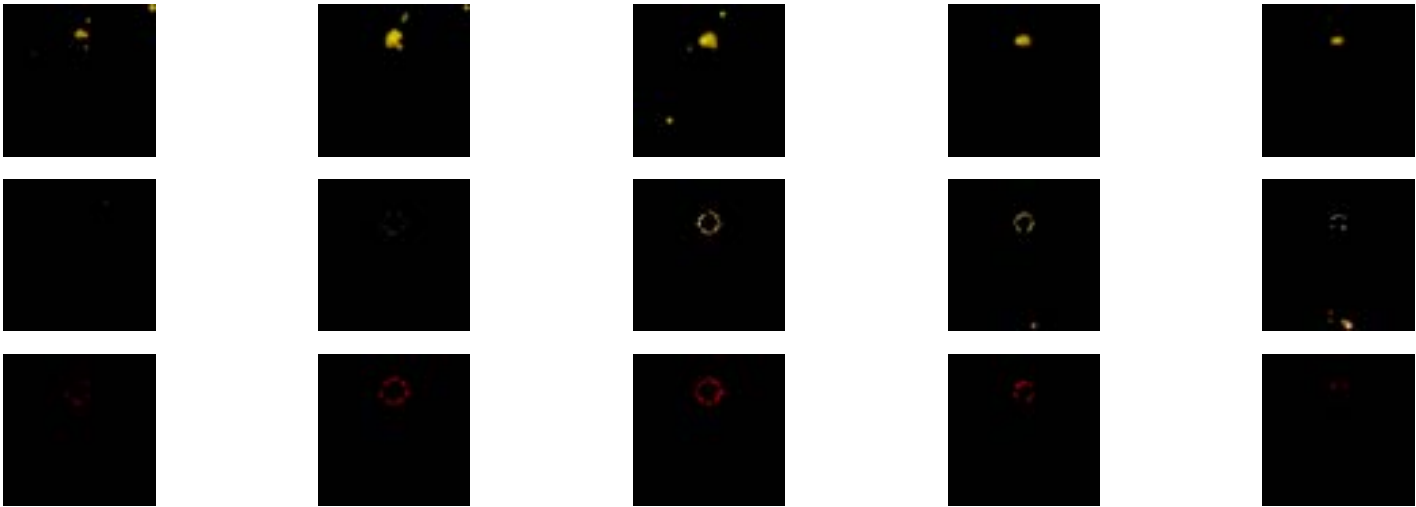


Figure 9: Images for the the z-slice visualizations. Rows in order is: validation_config, Medium, Large



bunny original view 1



bunny original view 2



bunny original view 3



bunny GT view 1



bunny GT view 2



bunny GT view 3



bunny GT view 1



bunny GT view 2



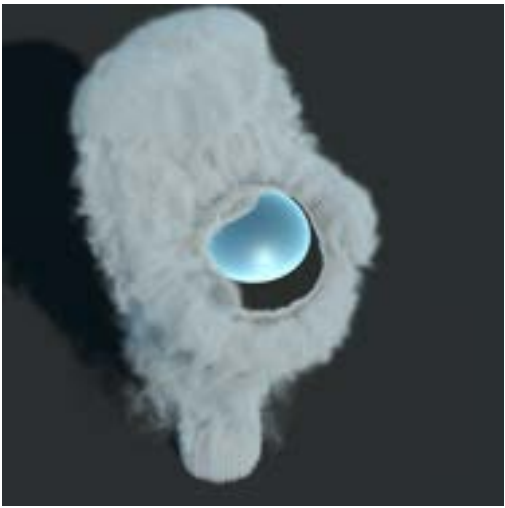
bunny GT view 3



plume original view 1



plume original view 2



plume original view 3



plume GT view 1



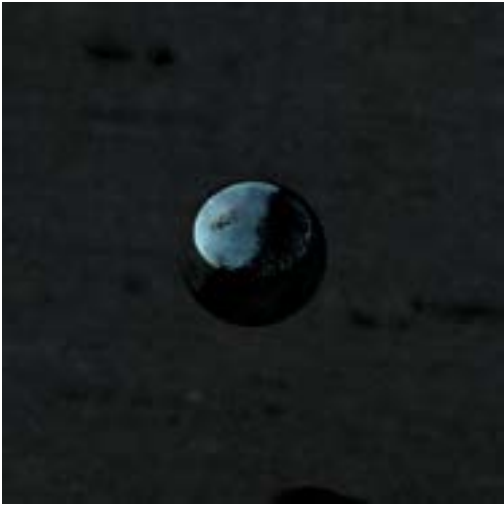
plume GT view 2



plume GT view 3



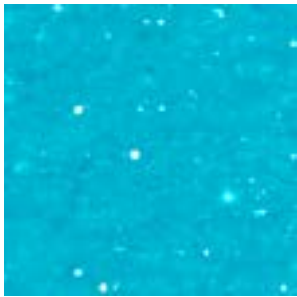
plume GT view 1



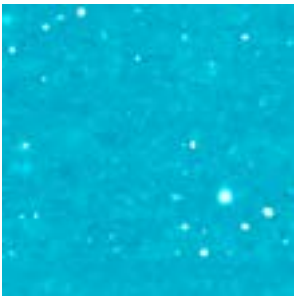
plume GT view 2



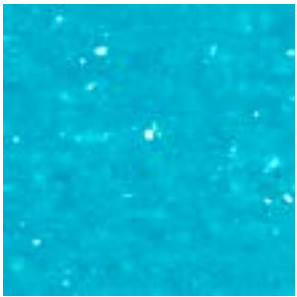
plume GT view 3



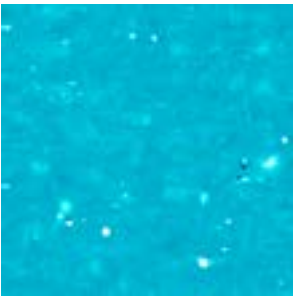
green_0_T



green_1_T



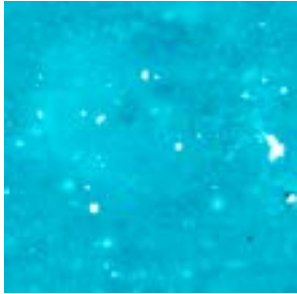
green_2_T



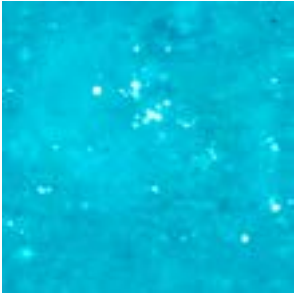
green_3_T



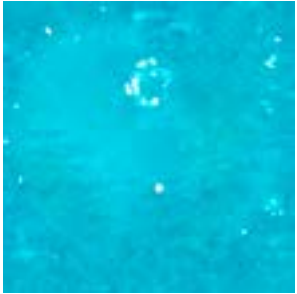
green_4_T



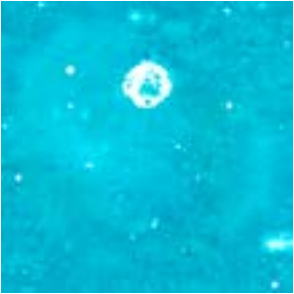
green_5_T



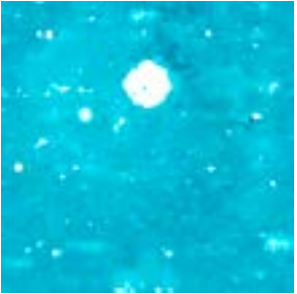
green_6_T



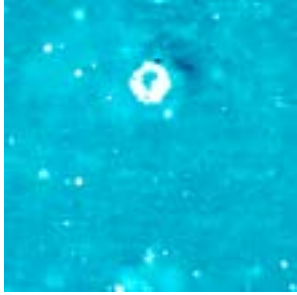
green_7_T



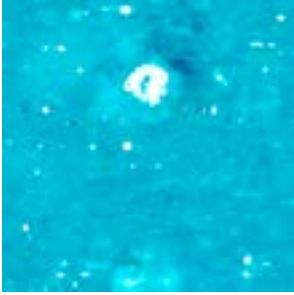
green_8_T



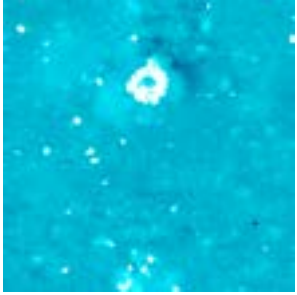
green_9_T



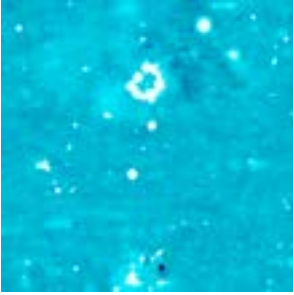
green_10_T



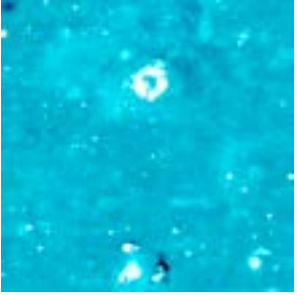
green_11_T



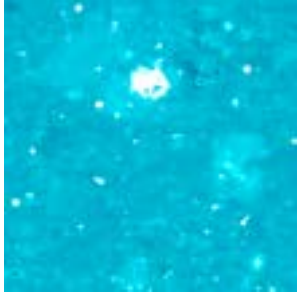
green_12_T



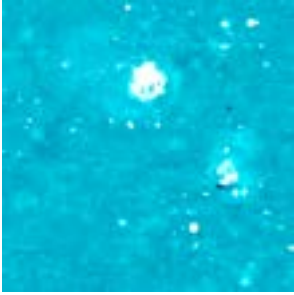
green_13_T



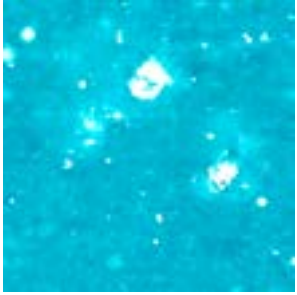
green_14_T



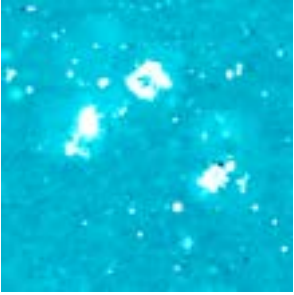
green_15_T



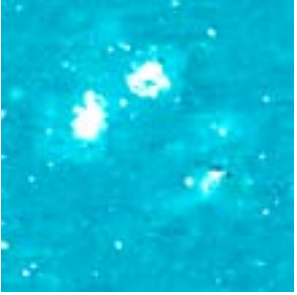
green_16_T



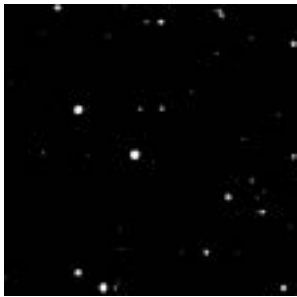
green_17_T



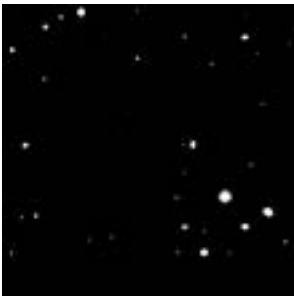
green_18_T



green_19_T



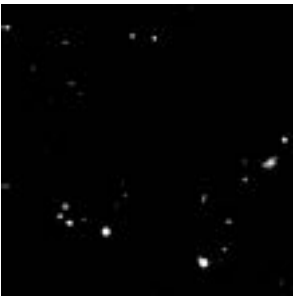
green_red_0_T



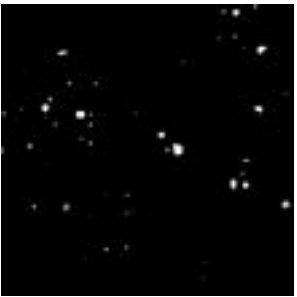
green_red_1_T



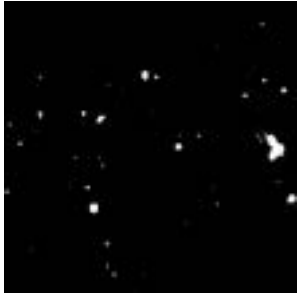
green_red_2_T



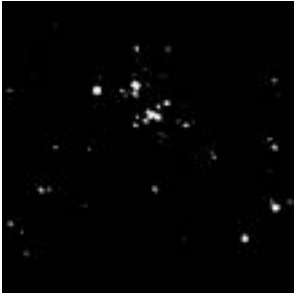
green_red_3_T



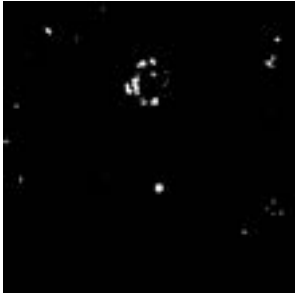
green_red_4_T



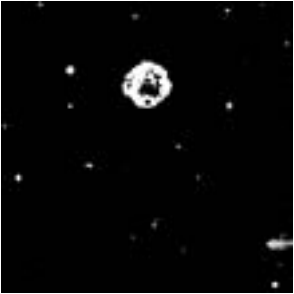
green_red_5_T



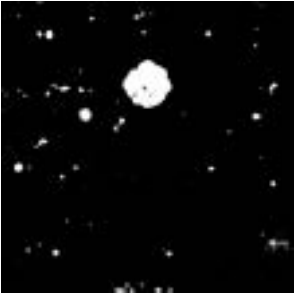
green_red_6_T



green_red_7_T



green_red_8_T



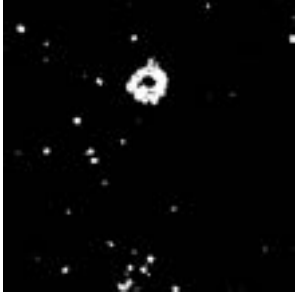
green_red_9_T



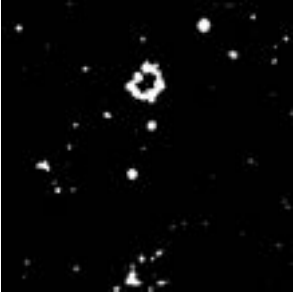
green_red_10_T



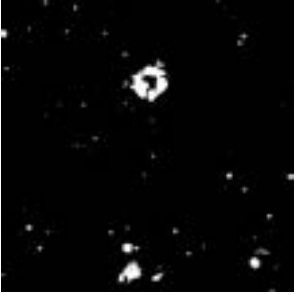
green_red_11_T



green_red_12_T



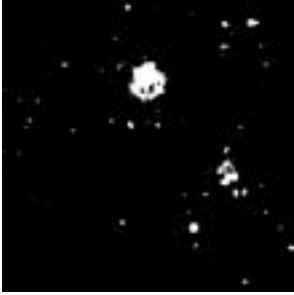
green_red_13_T



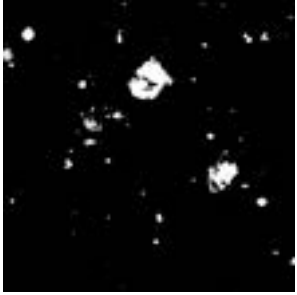
green_red_14_T



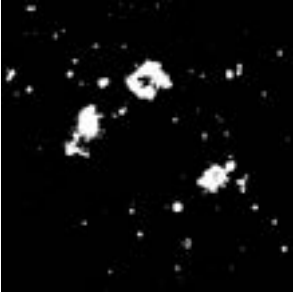
green_red_15_T



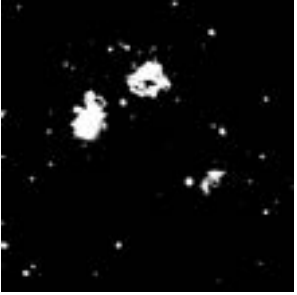
green_red_16_T



green_red_17_T



green_red_18_T



green_red_19_T



magenta_0_T



magenta_1_T



magenta_2_T



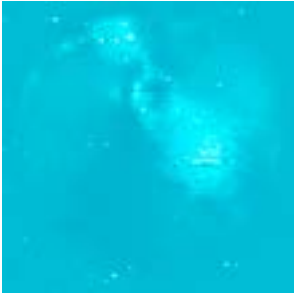
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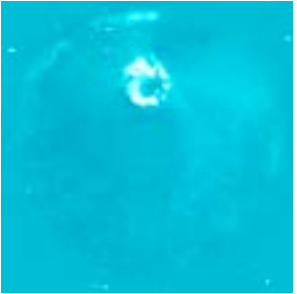
magenta_4_T



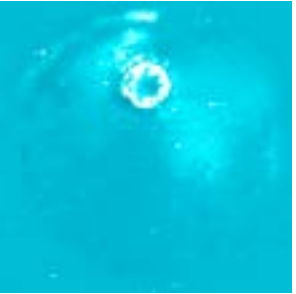
magenta_5_T



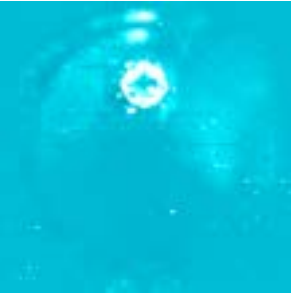
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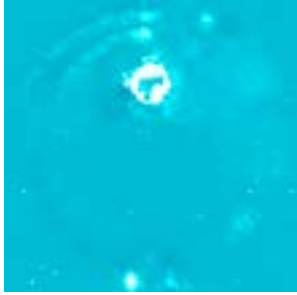
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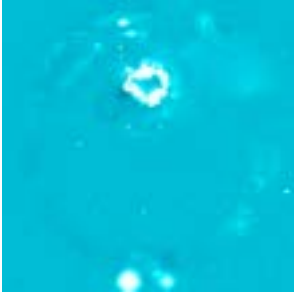
magenta_8_T



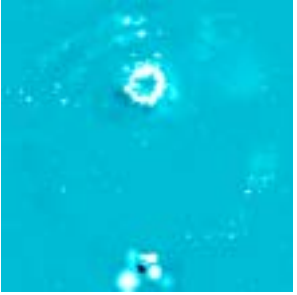
magenta_9_T



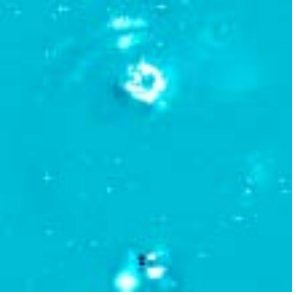
magenta_10_T



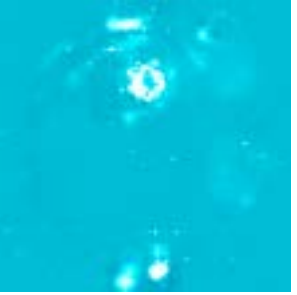
magenta_11_T



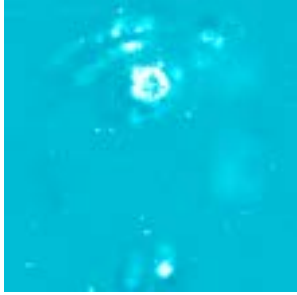
magenta_12_T



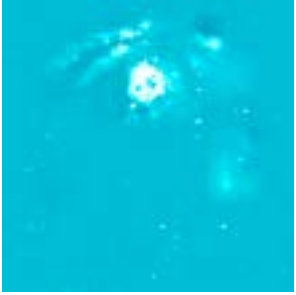
magenta_13_T



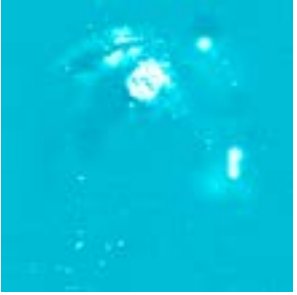
magenta_14_T



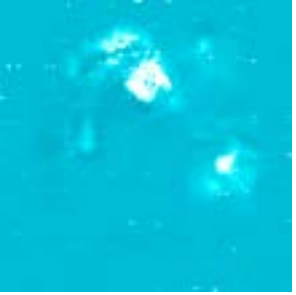
magenta_15_T



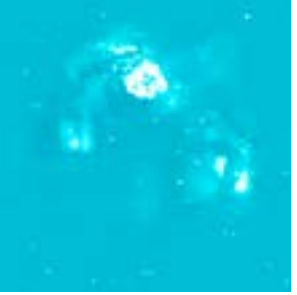
magenta_16_T



magenta_17_T



magenta_18_T



magenta_19_T



magenta_red_0_T



magenta_red_1_T



magenta_red_2_T



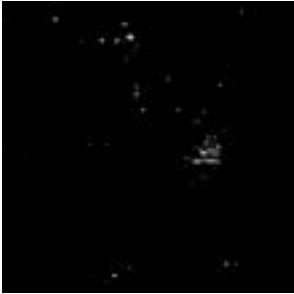
magenta_red_3_T



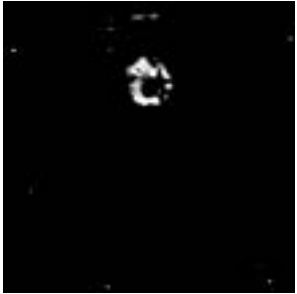
magenta_red_4_T



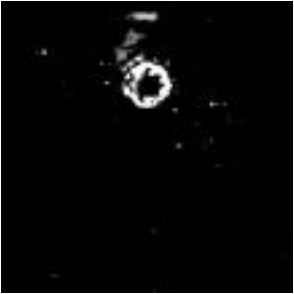
magenta_red_5_T



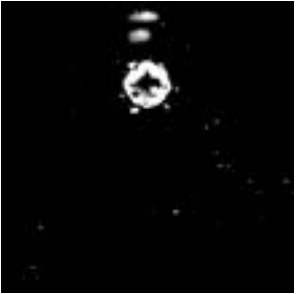
magenta_red_6_T



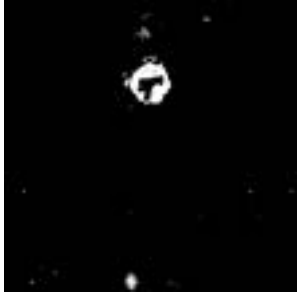
magenta_red_7_T



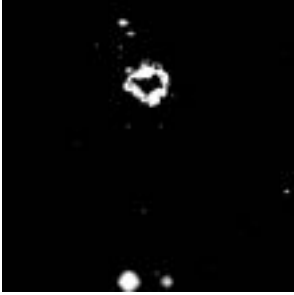
magenta_red_8_T



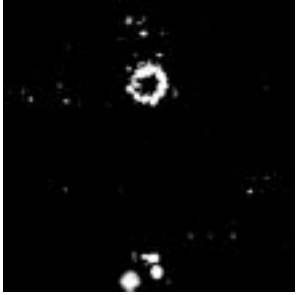
magenta_red_9_T



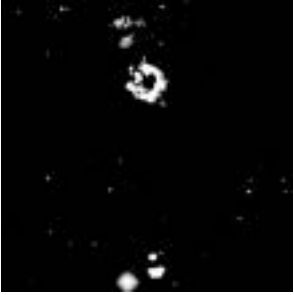
magenta_red_10_T



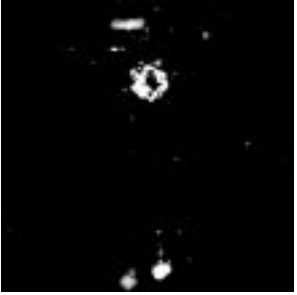
magenta_red_11_T



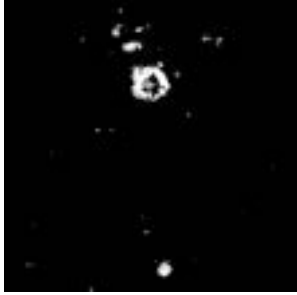
magenta_red_12_T



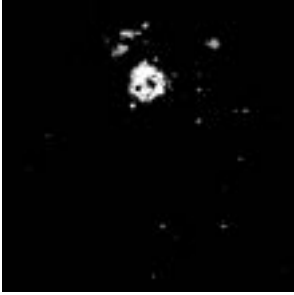
magenta_red_13_T



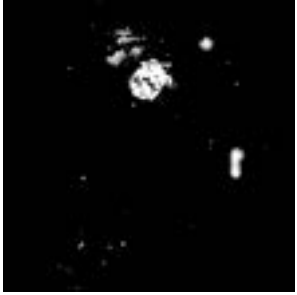
magenta_red_14_T



magenta_red_15_T



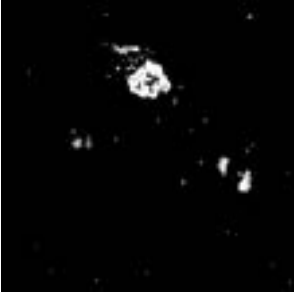
magenta_red_16_T



magenta_red_17_T



magenta_red_18_T



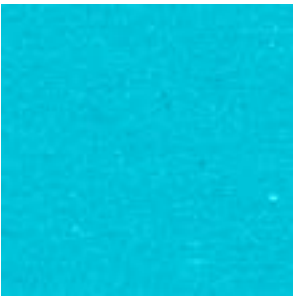
magenta_red_19_T



medium_0_T



medium_1_T



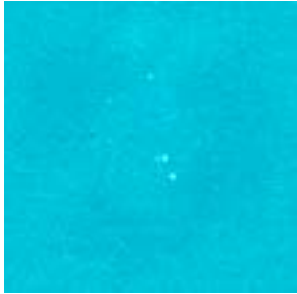
medium_2_T



medium_3_T



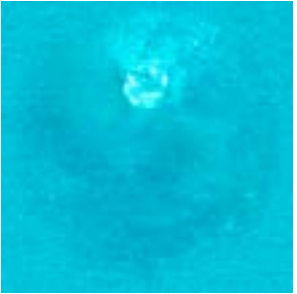
medium_4_T



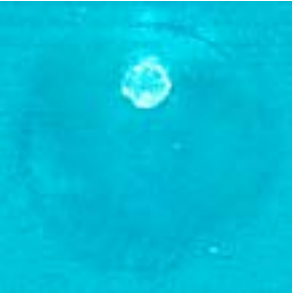
medium_5_T



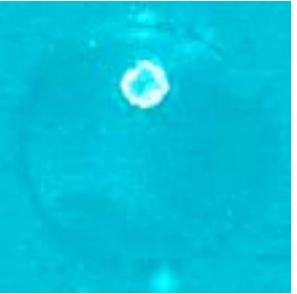
medium_6_T



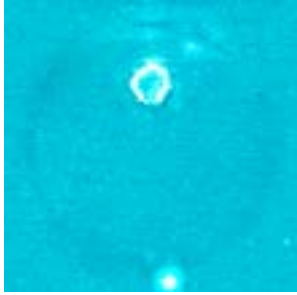
medium_7_T



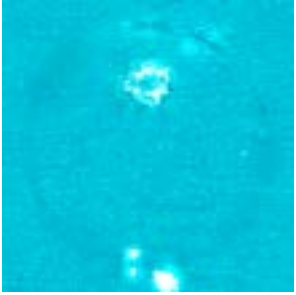
medium_8_T



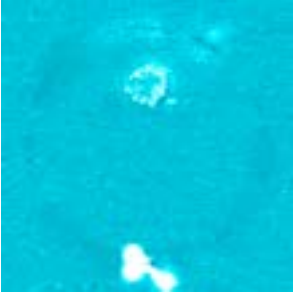
medium_9_T



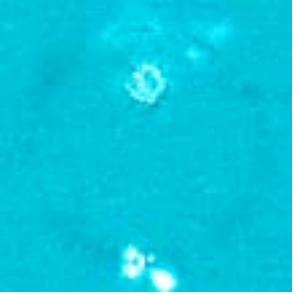
medium_10_T



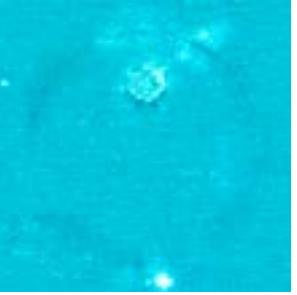
medium_11_T



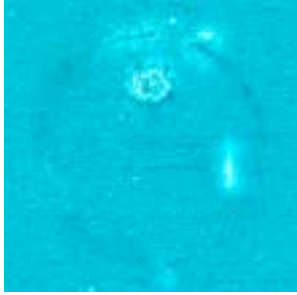
medium_12_T



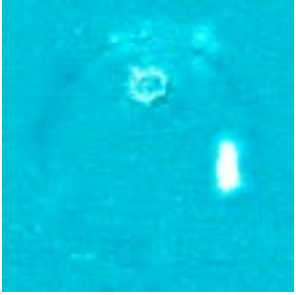
medium_13_T



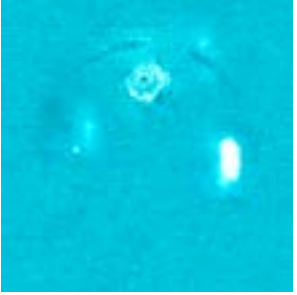
medium_14_T



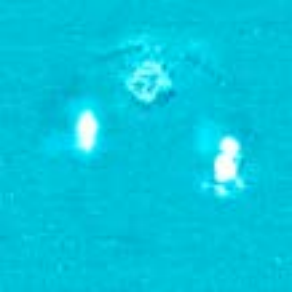
medium_15_T



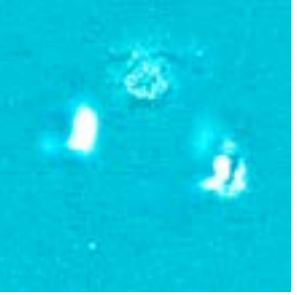
medium_16_T



medium_17_T



medium_18_T



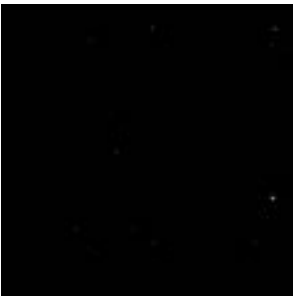
medium_19_T



medium_red_0_T



medium_red_1_T



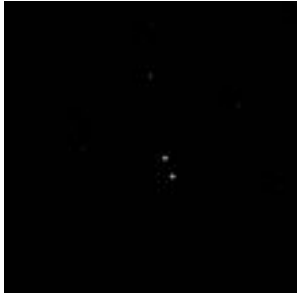
medium_red_2_T



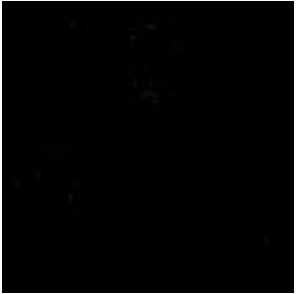
medium_red_3_T



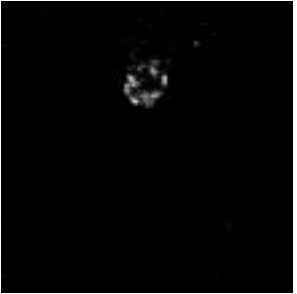
medium_red_4_T



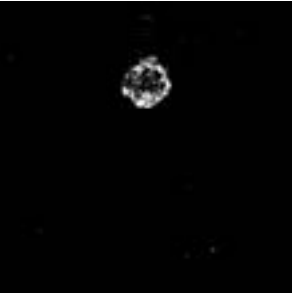
medium_red_5_T



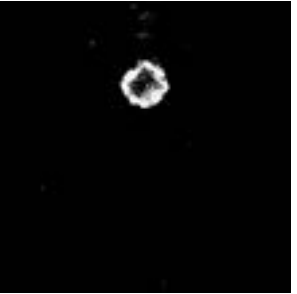
medium_red_6_T



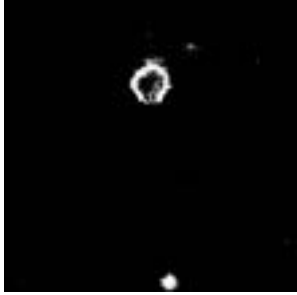
medium_red_7_T



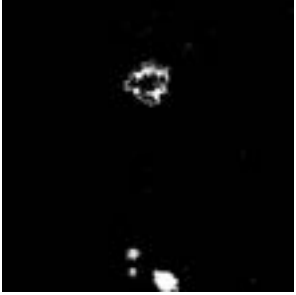
medium_red_8_T



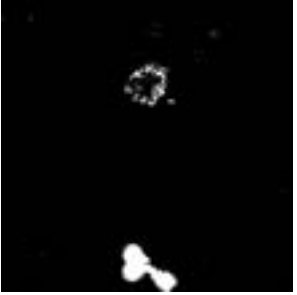
medium_red_9_T



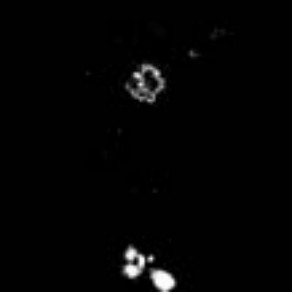
medium_red_10_T



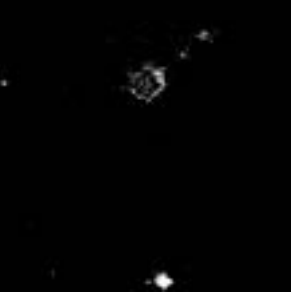
medium_red_11_T



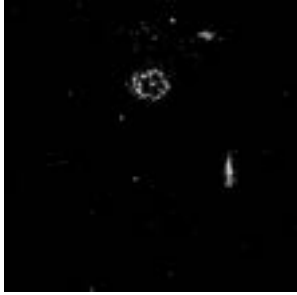
medium_red_12_T



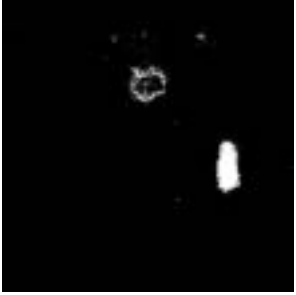
medium_red_13_T



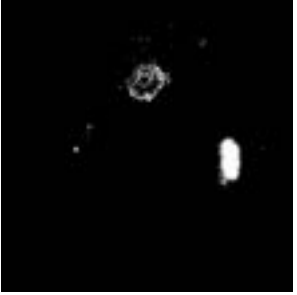
medium_red_14_T



medium_red_15_T



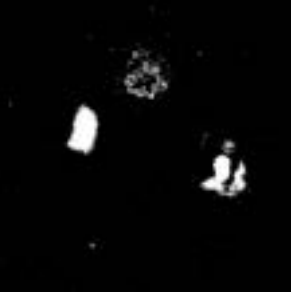
medium_red_16_T



medium_red_17_T



medium_red_18_T



medium_red_19_T



spectral_0_T



spectral_1_T



spectral_2_T



spectral_3_T



spectral_4_T



spectral_5_T



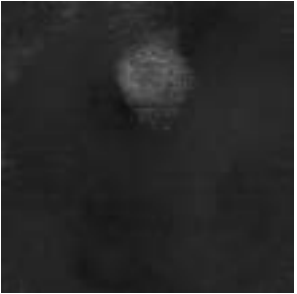
spectral_6_T



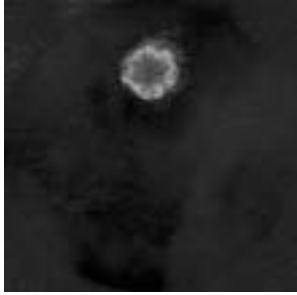
spectral_7_T



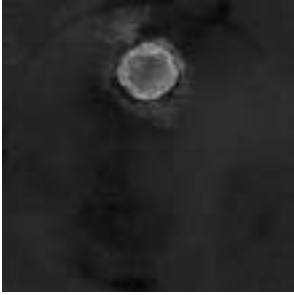
spectral_8_T



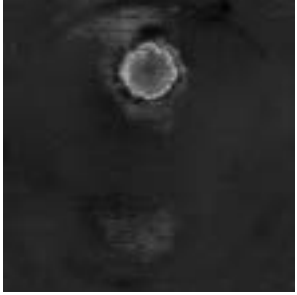
spectral_9_T



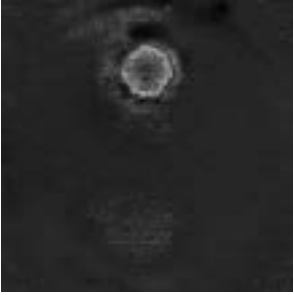
spectral_10_T



spectral_11_T



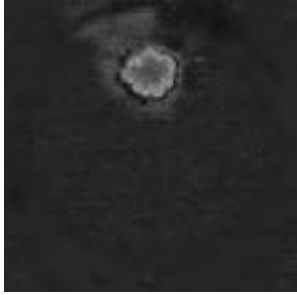
spectral_12_T



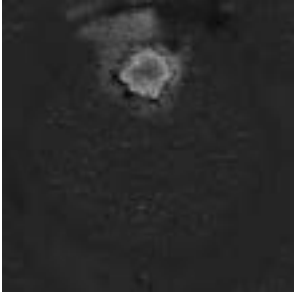
spectral_13_T



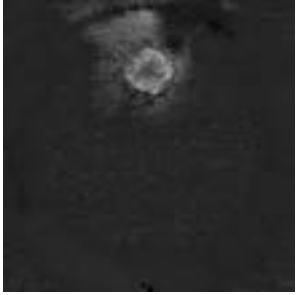
spectral_14_T



spectral_15_T



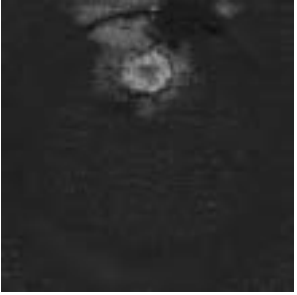
spectral_16_T



spectral_17_T



spectral_18_T



spectral_19_T



large_0_T



large_1_T



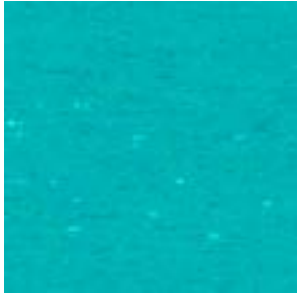
large_2_T



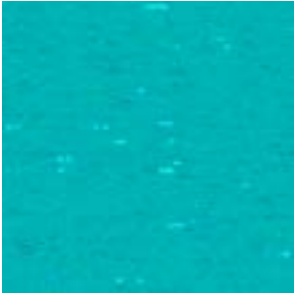
large_3_T



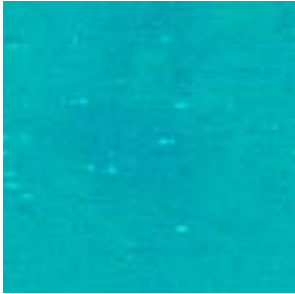
large_4_T



large_5_T



large_6_T



large_7_T



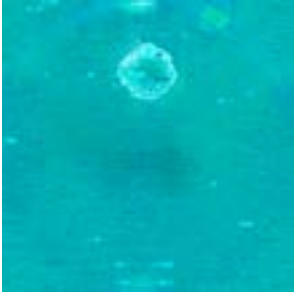
large_8_T



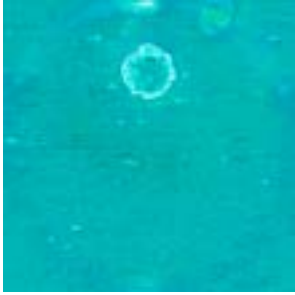
large_9_T



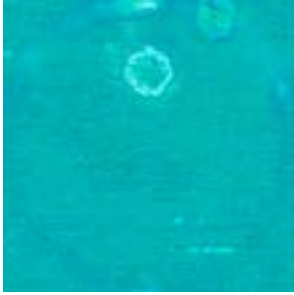
large_10_T



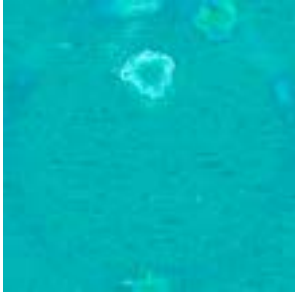
large_11_T



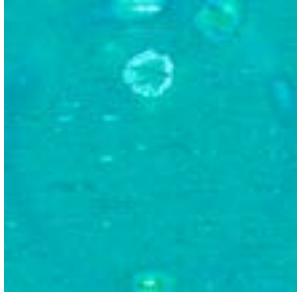
large_12_T



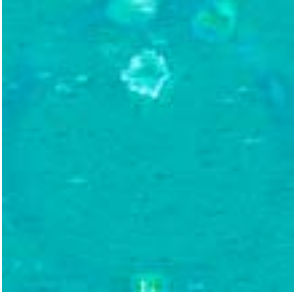
large_13_T



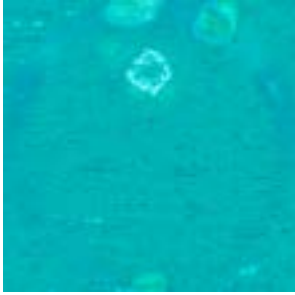
large_14_T



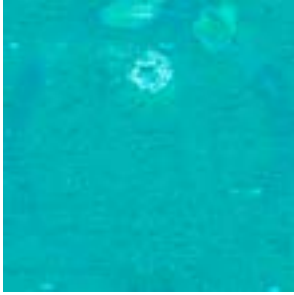
large_15_T



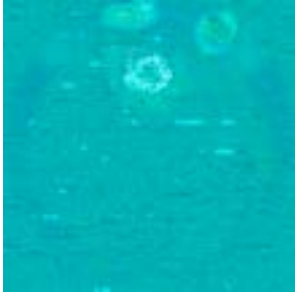
large_16_T



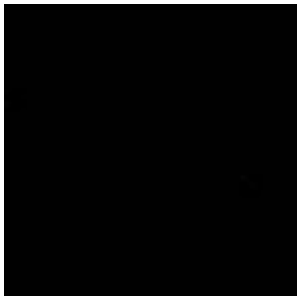
large_17_T



large_18_T



large_19_T



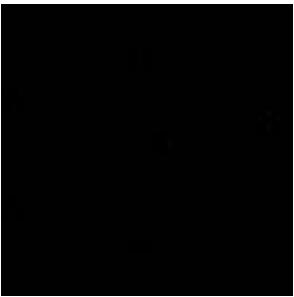
large_red_0_T



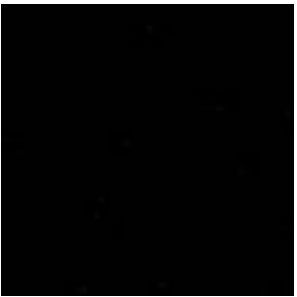
large_red_1_T



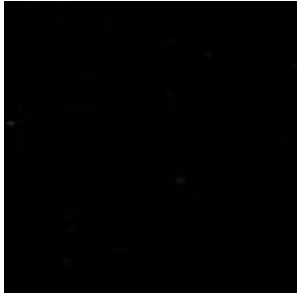
large_red_2_T



large_red_3_T



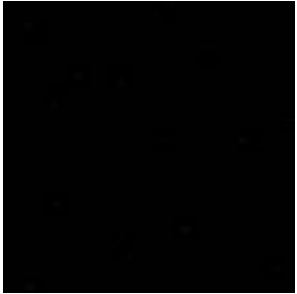
large_red_4_T



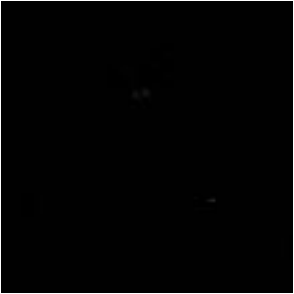
large_red_5_T



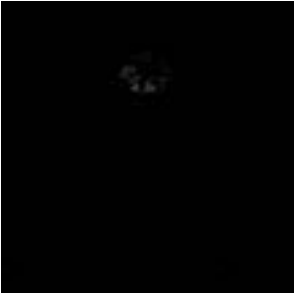
large_red_6_T



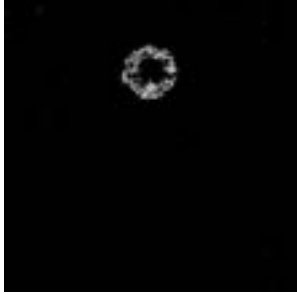
large_red_7_T



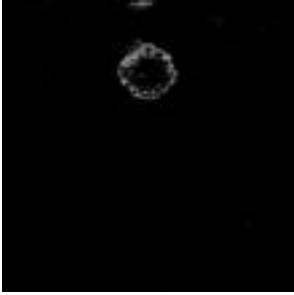
large_red_8_T



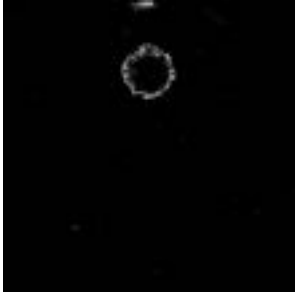
large_red_9_T



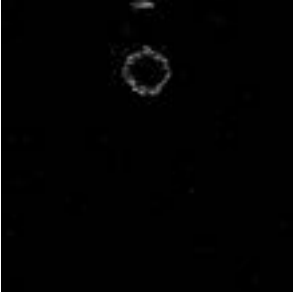
large_red_10_T



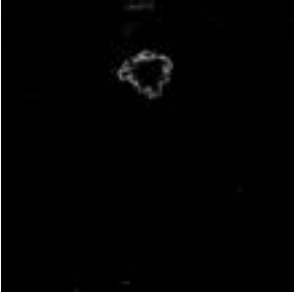
large_red_11_T



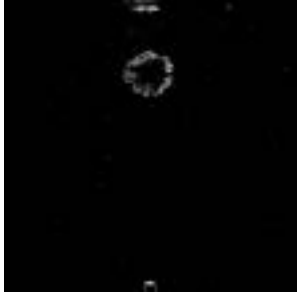
large_red_12_T



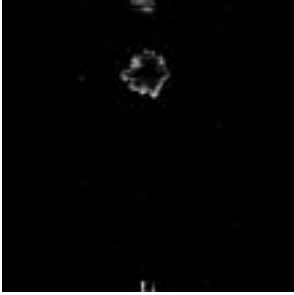
large_red_13_T



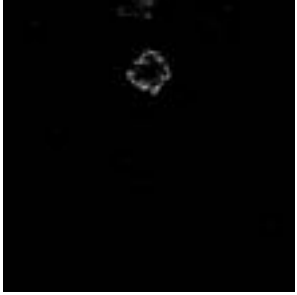
large_red_14_T



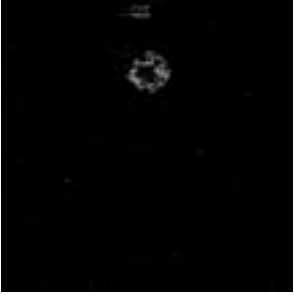
large_red_15_T



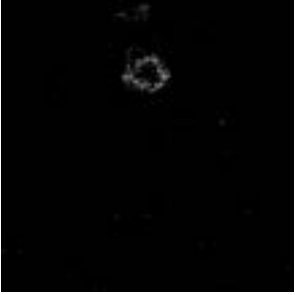
large_red_16_T



large_red_17_T



large_red_18_T



large_red_19_T

	mse	psnr (in dB)	final training loss	final training ema loss	training time (secs)
neurons128_layers4	0.348730	21.363102	0.347208	0.347263	208.521386
neurons128_layers3	0.070604	28.301794	0.046890	0.047271	194.315314
neurons128_layers2	0.068816	28.411854	0.046232	0.046520	171.721478
neurons64_layers4	0.069538	28.365958	0.045367	0.045371	160.505018
neurons64_layers3	0.070497	28.306432	0.044590	0.044862	152.979476
neurons64_layers2	0.068880	28.407300	0.044615	0.044698	142.817098
neurons32_layers4	0.070230	28.322791	0.044690	0.044656	136.415358
neurons32_layers3	0.069811	28.348950	0.044824	0.044848	136.516486
neurons32_layers2	0.070102	28.338299	0.045230	0.045251	127.072215

Table 3: Neurons and Layers validation testing results table

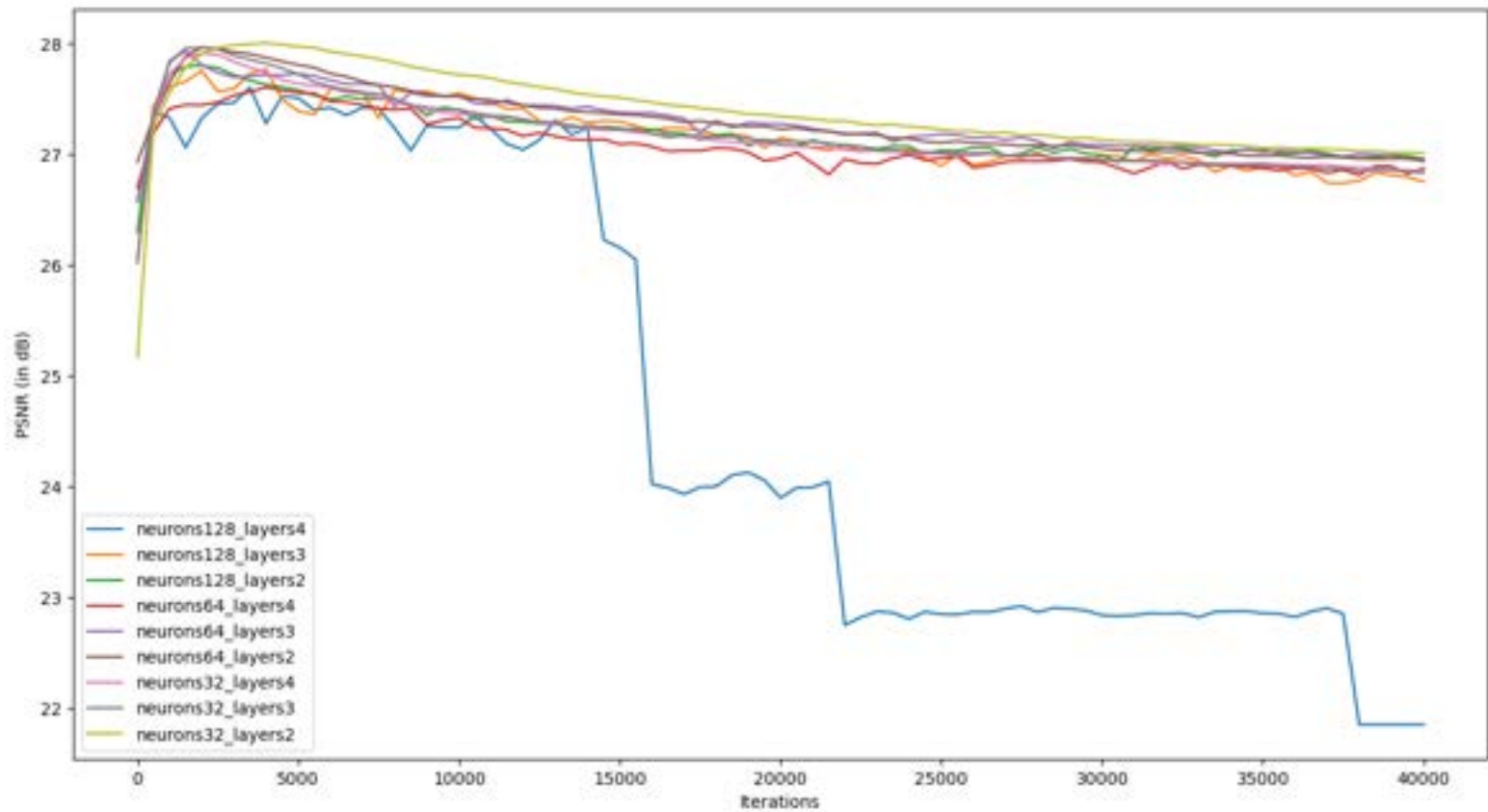


Figure 1: Graph of PSNR for Neurons and Layers validation testing

	mse	psnr (in dB)	final training loss	final training ema loss	training time (secs)
nlevels20	0.073566	28.121417	0.040660	0.040667	162.912744
nlevels16	0.069275	28.382394	0.044782	0.044714	143.221110
nlevels12	0.067233	28.512560	0.047723	0.047722	117.433949
nlevels8	0.062566	28.824698	0.050259	0.050320	94.906255
nlevels6	0.061805	28.878046	0.050444	0.050509	87.747340
nlevels4	0.060478	28.972330	0.050379	0.050439	72.690616
nlevels2	0.059532	29.040596	0.050811	0.050735	63.972096
nlevels1	0.062378	28.838082	0.050986	0.051023	55.449345

Table 4: n_levels validation testing results table

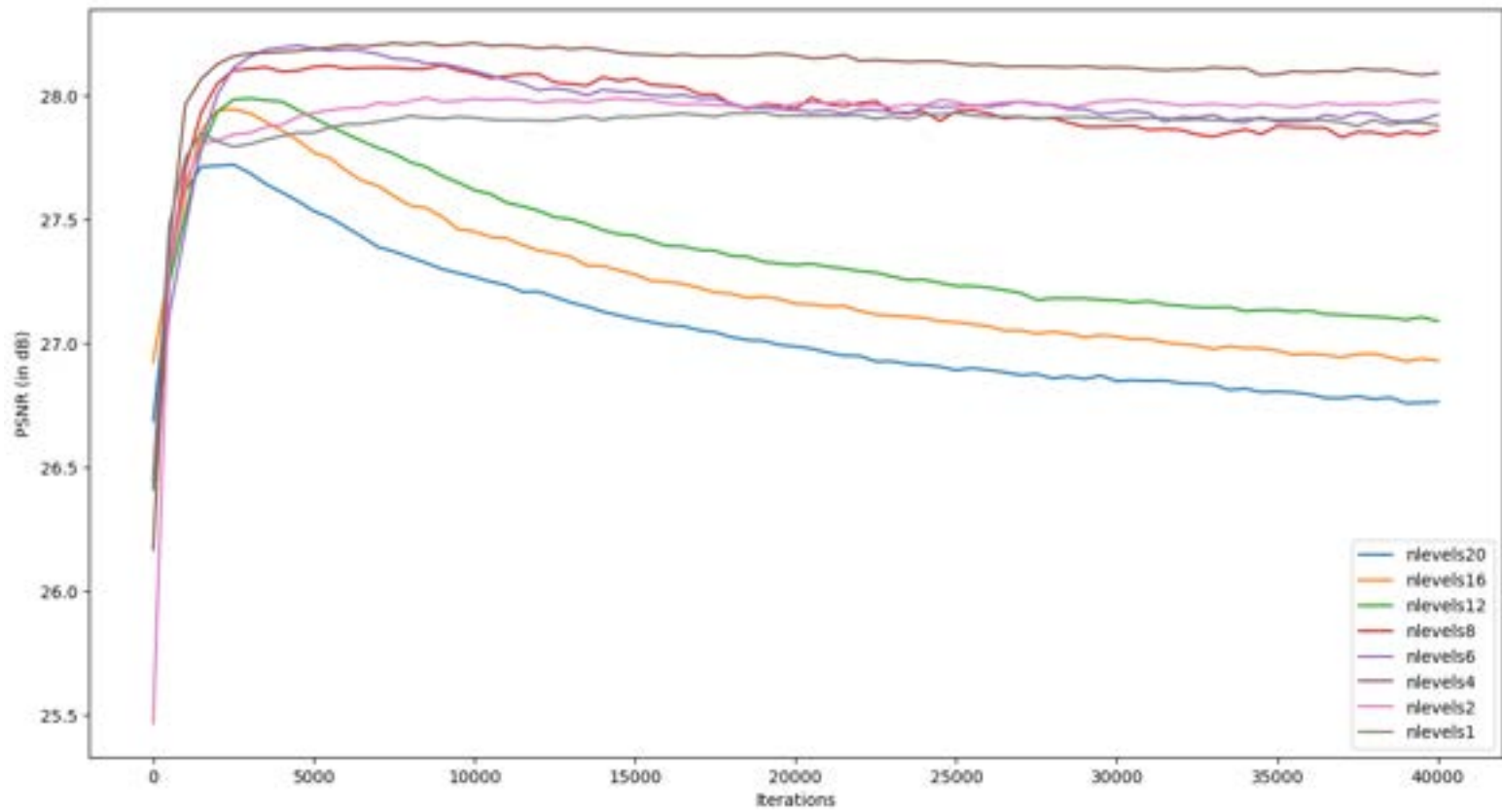


Figure 2: Graph of PSNR for n_levels validation testing

	mse	psnr (in dB)	final training loss	final training ema loss	training time (secs)
hashmapsize21	0.079420	27.789206	0.036752	0.036788	206.240166
hashmapsize19	0.068825	28.410693	0.044619	0.044669	143.211022
hashmapsize17	0.063777	28.741590	0.048727	0.048743	124.393271
hashmapsize15	0.061280	28.915052	0.050115	0.050186	112.220381
hashmapsize13	0.060603	28.963340	0.050234	0.050279	115.001244
hashmapsize11	0.059539	29.040277	0.050560	0.050503	175.035312
hashmapsize9	0.060128	28.997415	0.050539	0.050559	289.633266

Table 5: log_hashmapsize validation testing results table

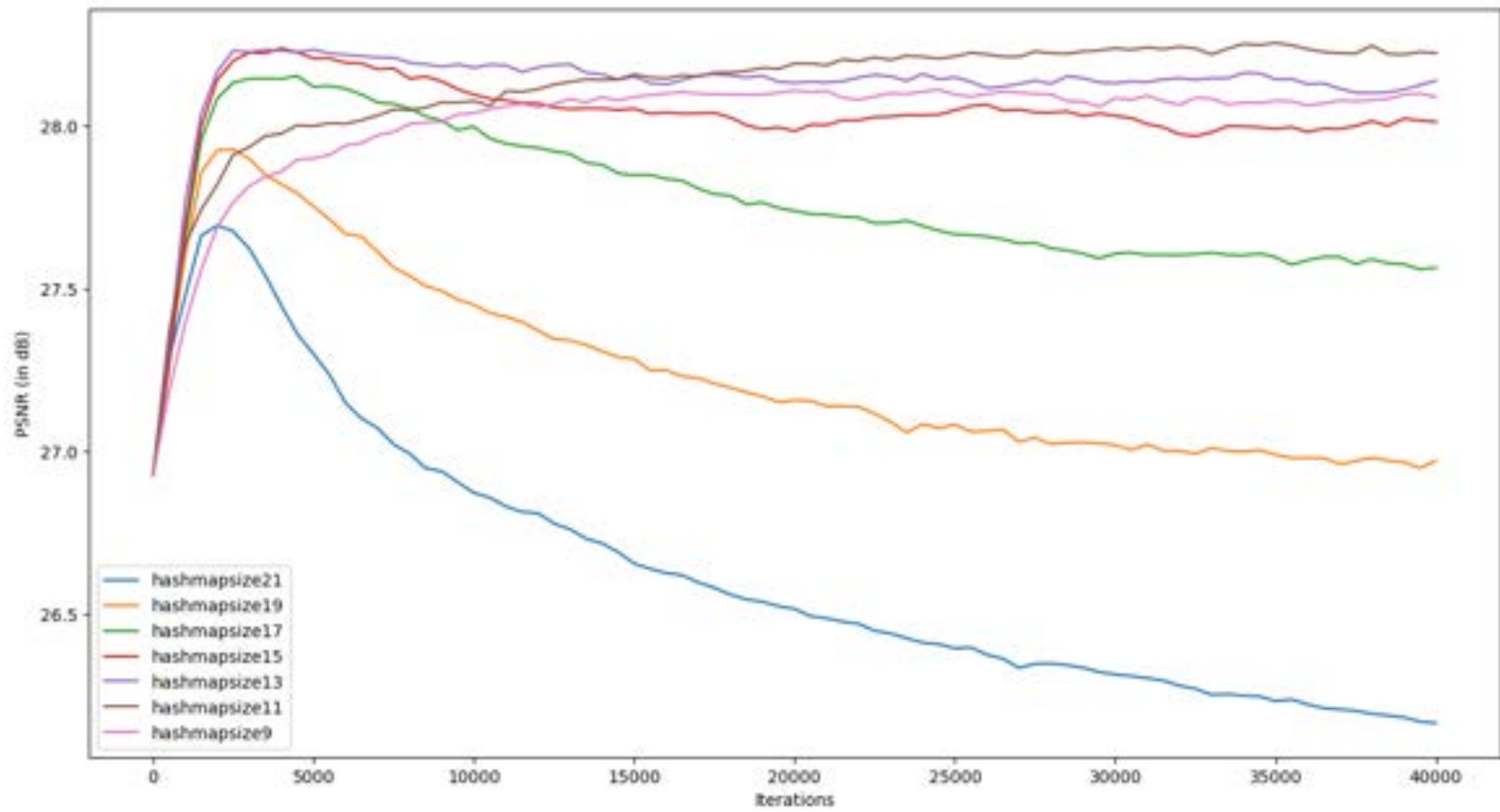


Figure 3: Graph of PSNR for log_hashmapsize validation testing

	mse	psnr (in dB)	final training loss	final training ema loss	training time (secs)
baseres32	0.070330	28.316856	0.043158	0.043134	134.912278
baseres24	0.069866	28.345595	0.043722	0.043745	138.168576
baseres16	0.069415	28.373727	0.044609	0.044667	142.268972
baseres8	0.068808	28.411767	0.046001	0.045986	159.401134

Table 6: baseres validation testing results table

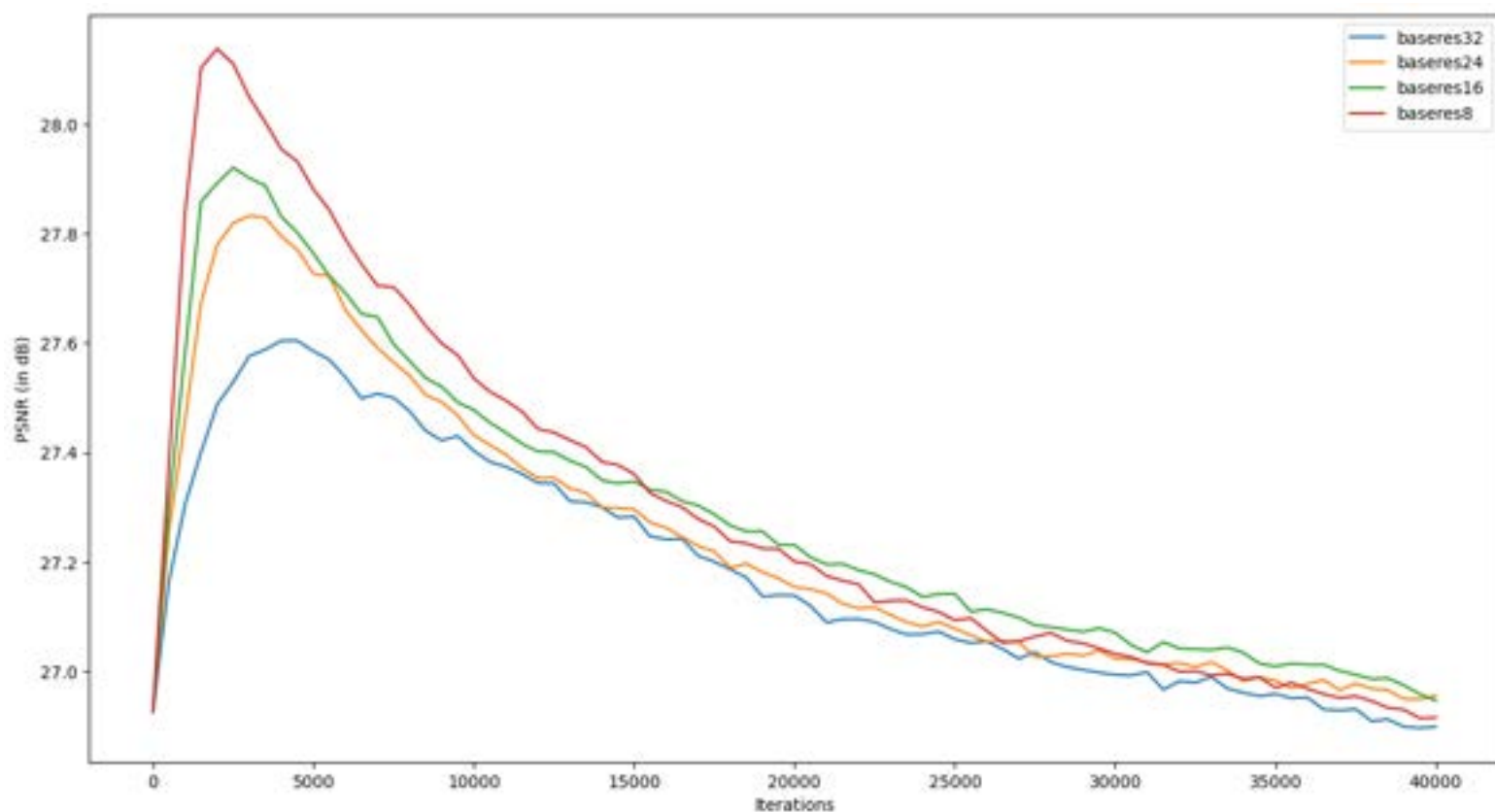


Figure 4: Graph of PSNR for baseres validation testing

	mse	psnr (in dB)	final training loss	final training ema loss	training time (secs)
shdegree6	0.068179	28.451541	0.044964	0.044916	143.617764
shdegree4	0.069362	28.376807	0.044713	0.044694	142.247992
shdegree2	0.068812	28.411508	0.045623	0.045901	138.371250

Table 7: shdegree validation testing results table

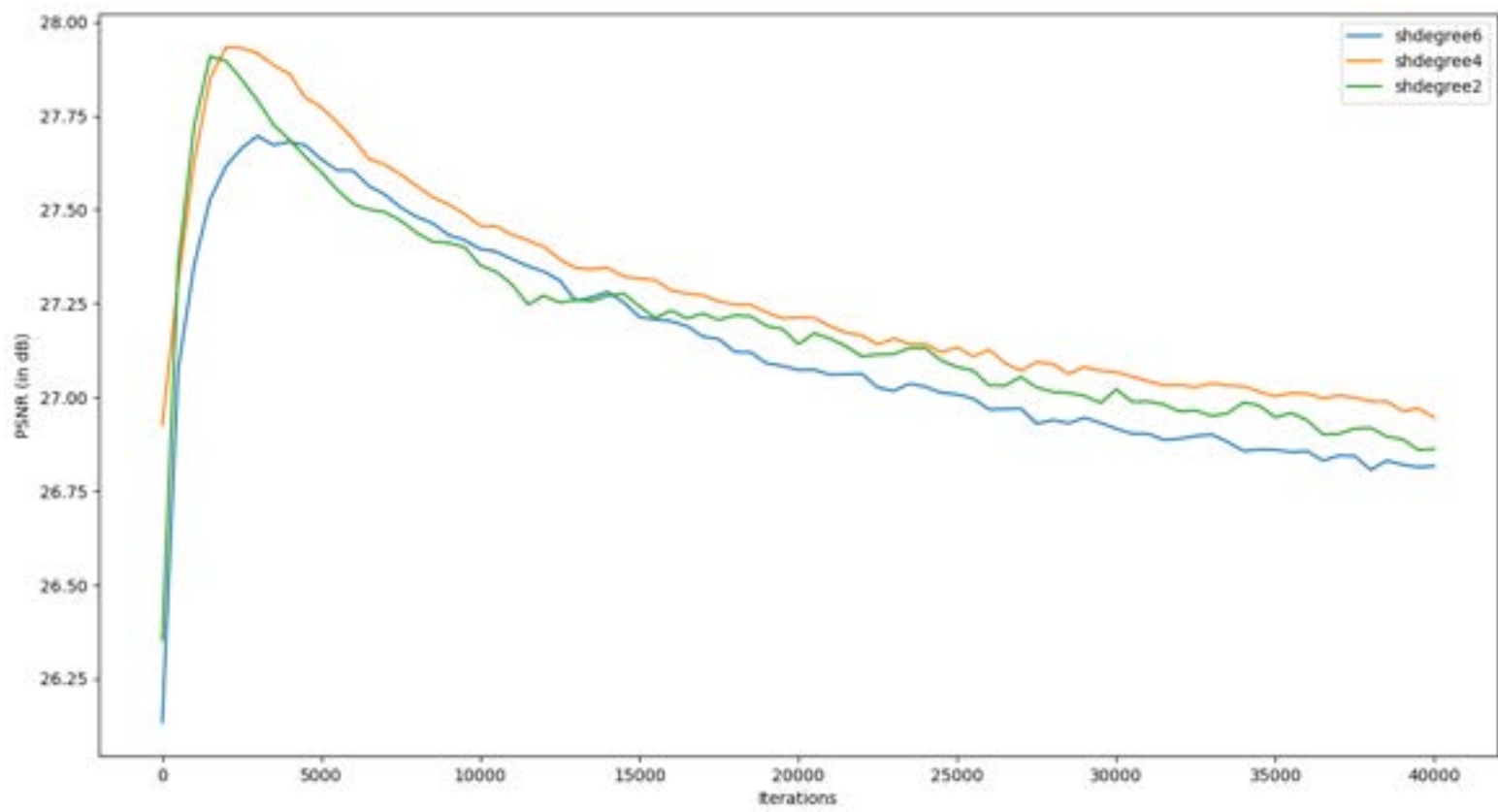


Figure 5: Graph of PSNR for shdegree validation testing

	mse	psnr (in dB)	final training loss	final training ema loss	training time (secs)
nfreq4	0.069615	28.361274	0.044619	0.044589	142.343500
nfreq6	0.069113	28.392750	0.044492	0.044512	142.583002
nfreq8	0.070818	28.286752	0.044377	0.044349	145.429194

Table 8: nfreq validation testing results table

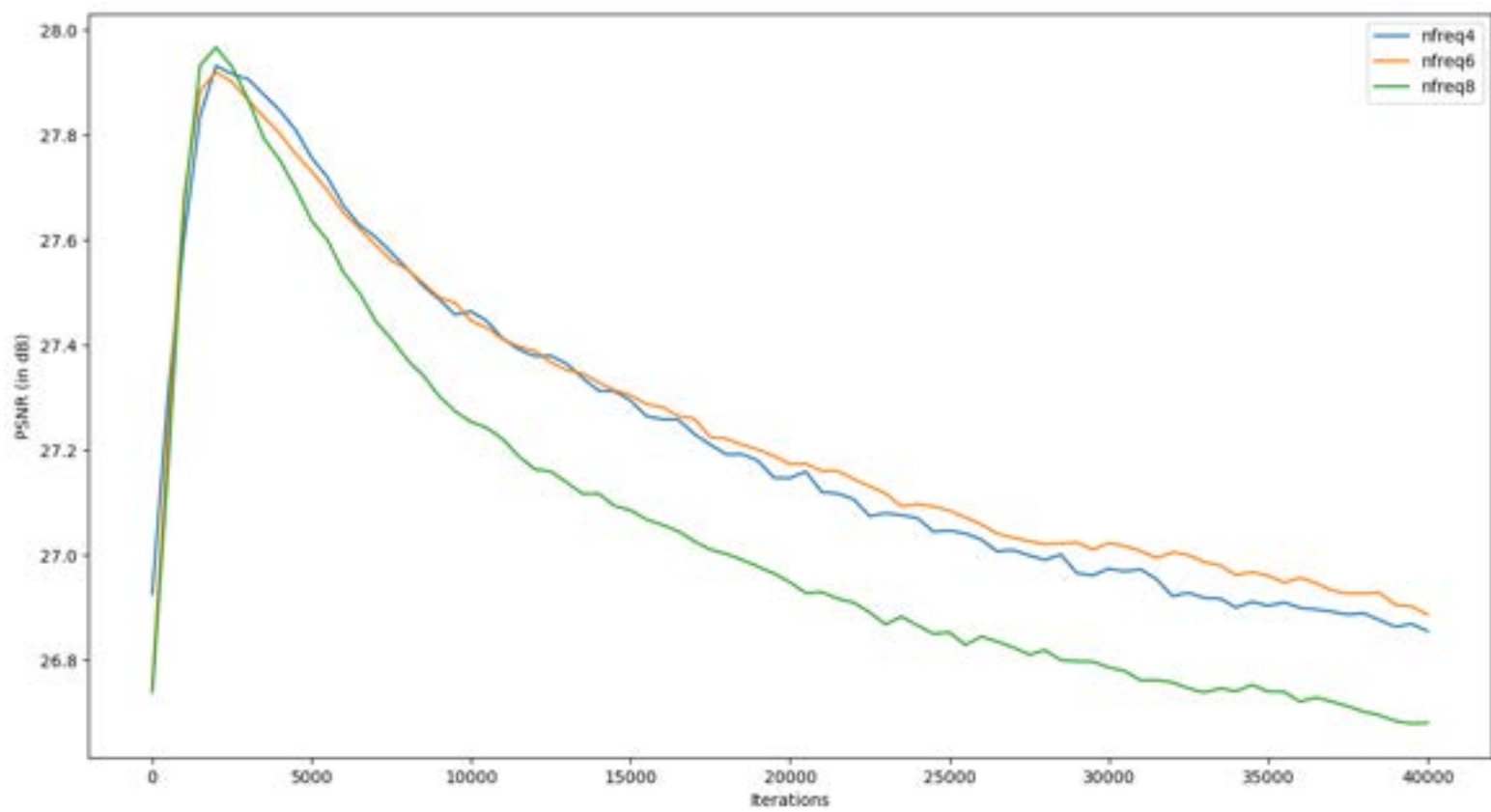


Figure 6: Graph of PSNR for nfreq validation testing

	mse	psnr (in dB)	final training loss	final training ema loss	training time (secs)
lr1e-1	0.070093	28.340064	0.054285	0.054425	136.561448
lr5e-2	0.075088	28.101344	0.061415	0.062196	140.162114
lr1e-2	0.068689	28.419978	0.047643	0.048304	142.129132
lr5e-3	0.074171	28.117863	0.052320	0.052192	142.344174
lr1e-3	0.069110	28.392673	0.044786	0.044829	143.222816
lr5e-4	0.067123	28.519436	0.045399	0.045361	143.403490
lr1e-4	0.064082	28.720737	0.048000	0.047922	143.614164

Table 9: lr validation testing results table

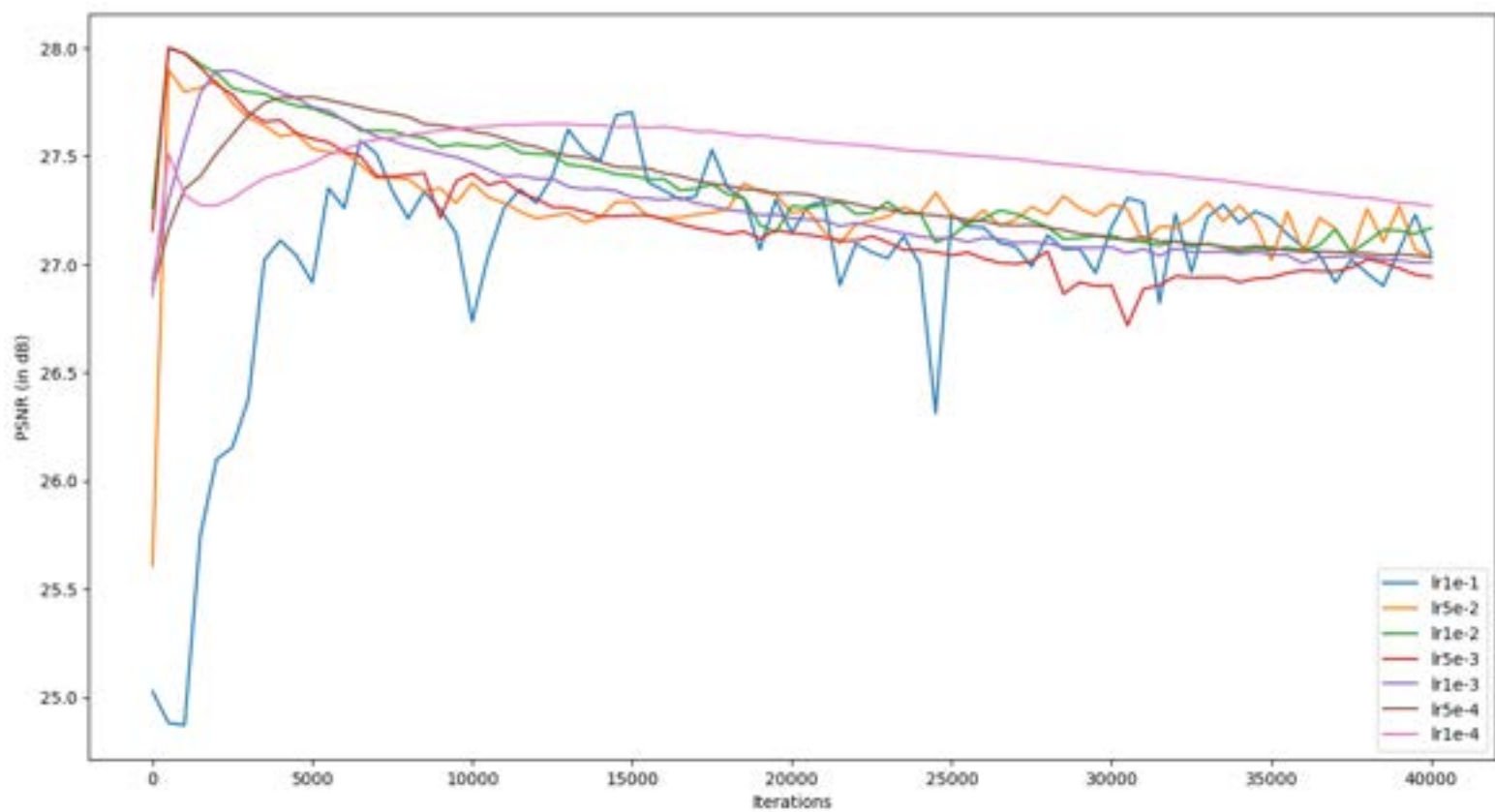


Figure 7: Graph of PSNR for lr validation testing